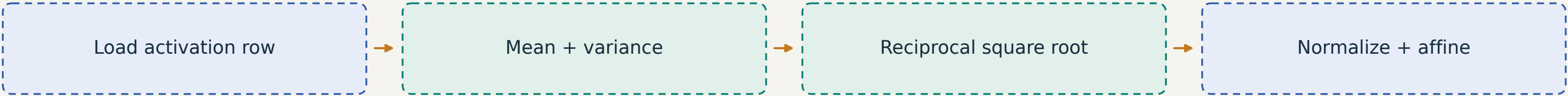


LayerNorm: reduce a row, then reuse its statistics

Standalone normalization · portable Triton historical comparison [N0, N1]

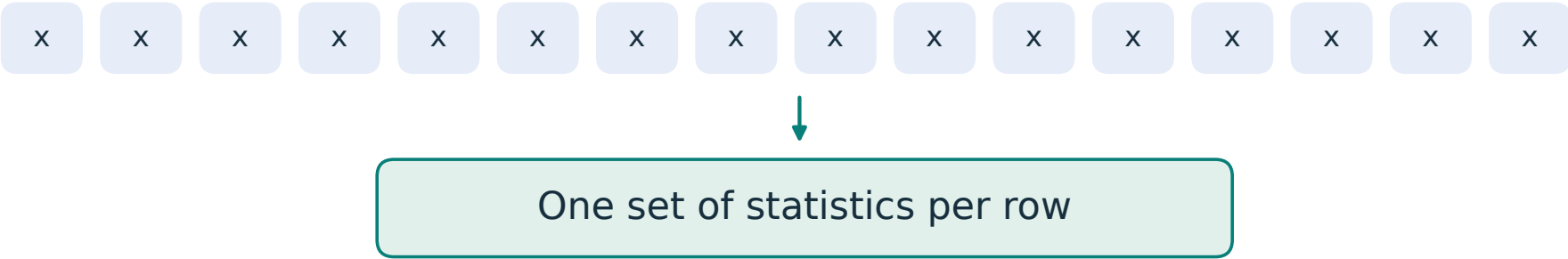
A / EXECUTION AND FUSION



Boxes show the row algorithm. Backward parameter reductions can require additional kernels.

B / ALGORITHM AND DATA REUSE

$$y_i = (x_i - \mu) (\sigma^2 + \varepsilon)^{-1/2} \gamma_i + \beta_i$$



Centering and scaling share row statistics.

HISTORICAL MEASUREMENT

Directly measured full operator F+B

L384 1.12×
0.276 → 0.246 ms

L768 1.09×
1.035 → 0.949 ms

Baseline: Pre-change portable Triton
Shape: D384

Measured gains concern the standalone portable path; they do not establish cross-operator CUDA fusion speedups.